The Summit Sabotage

Whodunnit? · Linear Equations & Slope Name: _____

The Snowpeak Winter Games were about to begin when the starter pistol was fired and nobody could find the **Golden Ski Pole**, the prize for the downhill champion. It had been on its stand in the lodge only minutes before!

Six people were at the resort, each in a different spot, and there are six ways to sneak a trophy off a mountain. Climb aboard, detective: **who** took it, **where** were they, and **how** did it get away?

How to crack the case

1. Your teacher will post **10 clue cards** around the room. Each card has a math problem and four answer choices. Every choice is paired with a clue.
2. Solve the problem on the **Show Your Work** page. Find your answer among the choices and read the clue next to it. If your answer is not there, check your work.
3. Use the clue to cross out suspects, places or methods on the case board below. Write the card number beside what it cleared.
4. After all 10 cards, exactly one suspect, one place and one method are left. Those three boxes name the culprit. If you end up with none, or more than one, a clue went wrong: recheck your math!

The Suspects	The Places	The Methods
☐ Coach Petra Ski Instructor	☐ Chairlift Station	☐ Stuffed in a ski bag
☐ Mr. Lund Lift Operator	☐ Ski Rental Shop	☐ Carried on a snowmobile
☐ Ms. Ferrari Racing Champion	☐ Summit Hut	☐ Dragged on a sled
☐ Hank Moreau Ski Patrol	☐ Lodge Café	☐ Hidden in a snowman
☐ Ms. Yamada Lodge Manager	☐ First Aid Cabin	☐ Tucked in a gondola seat
☐ Dr. Boone Weather Expert	☐ Tubing Hill	☐ Strapped to a snowboard

The culprit is _____ in _____ using _____

Show Your Work

The Summit Sabotage · Show every step in the matching box. Name: _____

1. Choice _____	**2.** Choice _____
3. Choice _____	**4.** Choice _____
5. Choice _____	**6.** Choice _____
7. Choice _____	**8.** Choice _____
9. Choice _____	**10.** Choice _____

Clue #1

The Summit Sabotage

Write the equation of the line through (0, 3) and (2, 7) in slope-intercept form.

A. $y = 2x + 3$	Coach Petra was fitting boots in the Ski Rental Shop. Six customers remember her.
B. $y = 4x + 3$	Coach Petra was resting in the Summit Hut after a morning lesson. Her students waved from the window.
C. $y = 3x + 2$	Dr. Boone was warming up with tea in the Lodge Café. The barista remembers the extra honey.
D. $y = -2x + 3$	Every sled was lined up and inspected. Nothing was dragged on any of them.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #2

The Summit Sabotage

Write the equation of the line with slope -2 that passes through (1, 4).

A. $y = 2x + 6$	The sled tracks in the snow all lead to the kids' hill, so no sled hauled the Pole away.
B. $y = -2x + 2$	Mr. Lund rode up to the Summit Hut to fix a heater. The repair ticket has his signature.
C. $y = -2x + 6$	Every ski bag was checked at the lodge door. None held the Pole.
D. $y = -2x + 4$	Dr. Boone was reading the wind gauge at the Chairlift Station. The weather log has his notes.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

Clue #3

The Summit Sabotage

Write the equation of the line through $(-2, 0)$ and $(2, 2)$.

A. $y = \frac{1}{2}x + 1$	Mr. Lund was running the Chairlift Station. The lift log shows him at the controls.
B. $y = 2x + 1$	Dr. Boone was having his gloves checked in the First Aid Cabin. The visitor book has his name.
C. $y = \frac{1}{2}x + 2$	Ms. Ferrari was signing autographs at the Summit Hut. A line of fans stretched out the door.
D. $y = -\frac{1}{2}x + 1$	The rescue sled has been locked in the shed since morning. It never carried the Pole.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #4

The Summit Sabotage

Write the equation of the line parallel to $y = 3x - 4$ that passes through $(0, 5)$.

A. $y = -\frac{1}{3}x + 5$	Hank Moreau was restocking the first-aid kit in the Summit Hut. The checklist is initialed.
B. $y = 3x - 4$	Dr. Boone was renting snowshoes at the Ski Rental Shop. The rental slip is stamped 9:06 a.m.
C. $y = 3x + 5$	The snowmobile’s cargo box is empty and covered in fresh snow. Nothing was carried.
D. $y = -3x + 5$	Fresh snow covers the sled path behind the lodge without a single mark.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

Clue #5

The Summit Sabotage

Write the equation of the line perpendicular to $y = 2x + 1$ that passes through $(4, -1)$.

A. $y = -\frac{1}{2}x + 1$	Ms. Ferrari was posing for photos on the Tubing Hill with a crowd of fans.
B. $y = -\frac{1}{2}x - 1$	Ms. Yamada was dropping off hot chocolate at the Summit Hut. The guestbook shows her visit.
C. $y = -2x + 7$	The sled rental tags show every sled came back before the pistol was fired.
D. $y = \frac{1}{2}x - 3$	Dr. Boone was giving a snow forecast on the Tubing Hill to a line of kids.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

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Clue #6

The Summit Sabotage

Write $4x - 2y = 6$ in slope-intercept form.

A. $y = -2x + 3$	A hiker’s photo shows Ms. Ferrari sipping cocoa in the Summit Hut as the pistol sounded.
B. $y = 2x - 6$	A ranger checked the sled rack and found only mittens.
C. $y = 2x + 3$	Photos show Dr. Boone at the Chairlift Station taking temperature readings right when the pistol fired.
D. $y = 2x - 3$	The snowman by the lodge was knocked over by kids. Only snow inside.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

Clue #7

The Summit Sabotage

A table of values has the points (0, −1), (1, 2) and (2, 5). Write the equation of the line.

A. $y = \frac{1}{3}x - 1$	Coach Petra was drawing a ski route on the chalkboard in the Summit Hut. Her drawing is still there.
B. $y = 3x - 1$	Hank Moreau was bandaging a skier's knee in the First Aid Cabin.
C. $y = -3x - 1$	A kid's plastic sled was the only one out that day, and it carried just a thermos.
D. $y = 3x + 1$	Dr. Boone was drawing a storm map on a napkin in the Lodge Café. Three skiers watched him.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

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Clue #8

The Summit Sabotage

A plant is 4 inches tall and grows 2 inches each week. Write an equation for its height y after x weeks.

A. $y = 4x + 2$	Dr. Boone was sliding down the Tubing Hill to test the snow. A ranger timed the run.
B. $y = 2x$	Hank Moreau was warming up in the Summit Hut after a patrol run. His radio log says so.
C. $y = 2x + 4$	The gondola seats were cleaned at 9 a.m. and nothing was left behind.
D. $y = -2x + 4$	The sled dogs stayed in the kennel all day, so no sled pulled anything off the mountain.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

Clue #9

The Summit Sabotage

A line crosses the x-axis at (6, 0) and the y-axis at (0, 3). Write its equation.

A. $y = -2x + 3$	Mr. Lund was eating lunch in the Summit Hut. Two skiers shared his table.
B. $y = -\frac{1}{2}x + 3$	Ms. Yamada was serving hot cocoa in the Lodge Café during the whole blackout.
C. $y = -\frac{1}{2}x + 6$	Sled runners leave deep grooves, and the snow near the exit is smooth.
D. $y = \frac{1}{2}x + 3$	Dr. Boone was waxing a pair of skis in the Ski Rental Shop. The mechanic vouches for him.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #10

The Summit Sabotage

Write the equation of the line through (−3, 2) and (5, 2).

A. $y = x + 2$	The sled by the lodge door was loaded with firewood only, and the crew unloaded it in front of everyone.
B. $y = 2x$	Ms. Yamada was hanging the new banner in the Summit Hut when the pistol went off.
C. $y = -2$	Dr. Boone was in the First Aid Cabin getting a bandage for a blister. The nurse’s log shows the time.
D. $y = 2$	All the snowboards were checked at the rack. None had the Pole strapped to them.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.